UdyamSync — solution architecture

One stateless API, deterministic rules for anything that touches price or compliance, an optional LLM only for language, and every identifier masked or encrypted.

WHO USES IT

Owner app (PWA)
Hindi · Marathi · English built in, 7 more via LLM · voice in and out on the device · resumes after a dropped connection

WhatsApp view
Same advisor session; quote card hands off to the web OneBill

Partner Co-Pilot
Win-back inbox, assisted mode on a tablet, commission credit, KYC exceptions

Signatory link
Remote e-sign on the signatory's own phone

Insights + KYC Inspector
Replay the Quarter, compliance report, specimen inspector

FASTAPI SERVICE

UdyamSync API

Python 3.11 · 65 tests in CI

Vyapar Saathi advisor
Rule NLU first, LLM fills gaps · 3-turn discovery · objections · human hand-off

Catalogue rules engine
City tier · team size · sites · active GSTIN · product dependencies

OneBill quote
Tiered pricing by term · setup waiver · GST · multi-site feasibility

Smart KYC
On-server OCR · deskew · watermark removal · 20+ registry checks · field boxes

Consent ledger
Per-purpose DPDP consent, hash-chained + HMAC, versioned notice

Vault + purge
Fernet encryption at rest · images purged 30 days after verification

Orders & tracker
5 stages · per-product ETA · WhatsApp updates · SLA watch · remote e-sign

Win-back model
Logistic propensity (AUC 0.73) × drop-reason recoverability · next best action

Analytics
Replay the Quarter · compliance re-run · segment matrix · live telemetry

Design rule: the AI talks, the rules engine decides. Prices and eligibility come only from the catalogue; any ₹ figure the model writes that is not in the quote is discarded. Orders are re-priced and re-checked on the server.

Journey: Discover → OneBill → Consent → Smart KYC → e-Sign + AutoPay → Track

PROTOTYPE
- SQLite JSON docs
- Fernet key from secret
- CSV registries
- Mocked DigiLocker, e-Sign, UPI, WhatsApp
- OCR in the API process

→

PRODUCTION
- Postgres + object storage
- KMS keys with rotation
- GSTN / Udyam via licensed GSP
- DigiLocker Requester, ASP e-Sign, NPCI AutoPay, Accel WhatsApp API
- OCR worker queue behind a load balancer

DATA & SERVICES

Case dataset
Catalogue (25) · 5,000 leads · GSTIN (3,534) and Udyam (2,723) registries · 47 KYC specimens

SQLite store → Postgres
Sessions, leads, orders, consents, append-only audit log

Encrypted vault
Raw GSTIN and document images only, never in logs

Gemini / Groq (optional)
Free tier, redacted input, never sees documents OPTIONAL

DigiLocker · e-Sign · UPI AutoPay · WhatsApp BSP · GSTN/Udyam APIs
Mocked in the prototype behind the same interfaces PRODUCTION SWAP